both sides of the South Atlantic, even though they were thousands of miles apart. In his 1912 theory of continental drift, Wegener cited this as evidence that continents were once joined and had broken away. It was not until the 1960s that a viable mechanism was found for such movement. Geophysicists discovered patterns of magnetic anomalies running in parallel stripes on either side of ocean ridges and identified the process of seafloor spreading—hot magma bubbling up through cracks in the oceanic crust and forming new crust as it cools and moves away. This gradual process shifts and shapes continents.

The birth of biogeography

In the Age of Exploration from the 16th century on, scientists began to study the geographical distribution of plants and animals. By the 1860s, Alfred Russel Wallace viewed these patterns, clearly defined by physical barriers such as mountains and seas as a key supporting argument for evolution. Wallace noted, for example, the ocean straits that produced a sharp division between the flora and fauna of Australasia and Southeast Asia.

With a better understanding of Earth's biogeography, 20th-century ecologists divided the planet into biomes—broad communities of flora and fauna that interact in different habitats, such as tropical rain forests or tundra. Botanist Leslie Holdridge refined the concept in 1947 with his life zone classification, in which he mapped zones based on the two crucial influences on vegetation: temperature and rainfall.

A "whole Earth" approach

The word "biosphere" was coined by Austrian geologist Edward Suess in 1875 to signify all the areas at or near the surface of the Earth where organic life can exist. In 1926, the Russian geochemist Vladimir Vernadsky